Multi-Floor Building Network Project Report

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Course: Computer Networks

Project: Multi-Floor Building Network with OSPF, VLANs, DHCP, DNS, and Web

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Project Objective

The purpose of this project was to design and implement a multi-floor building network in Cisco Packet Tracer with the following requirements:

1. Use three routers (one per floor) and three switches for interconnectivity.
2. Configure VLANs for each floor.
3. Implement inter-VLAN routing using Router-on-a-Stick configuration.
4. Configure DHCP servers on each floor to dynamically assign IP addresses to PCs.
5. Add a central DNS server and a web server to provide name resolution and website access.
6. Implement OSPF routing to dynamically manage communication between all networks.
7. Verify network functionality and troubleshoot potential link failures.

Network Devices Used

Device	Quantity	Notes
Router-PT (2911)	3	One per floor
Switch-PT (2960)	3	One per floor
PC-PT	9	3 per floor
Server-PT	2	DNS & Web servers
Copper Straight-Through Cable	Many	PC ↔ Switch & Server ↔ Switch
Serial DCE Cable (purple)	2	Router ↔ Router connections

Network Topology

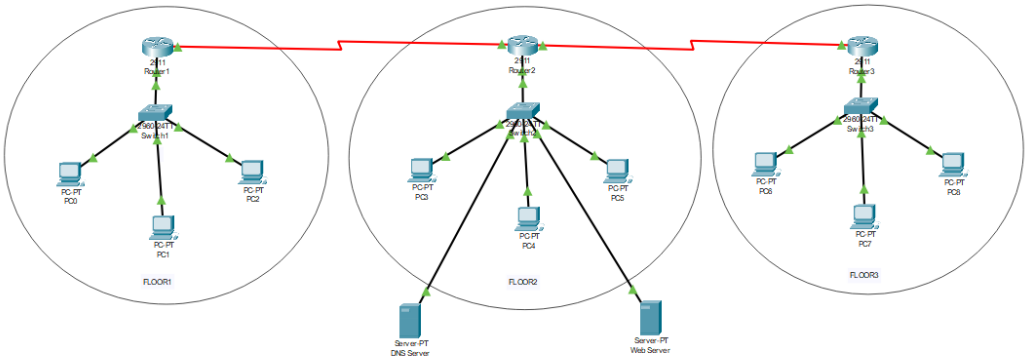
Floor 1: - PCs: PC1, PC2, PC3 → Switch1 → Router1 (trunk)

Floor 2: - PCs: PC4, PC5, PC6 → Switch2 → Router2 (trunk) - Servers: DNS-SERVER & WEB-SERVER → Switch2

Floor 3: - PCs: PC7, PC8, PC9 → Switch3 → Router3 (trunk)

Router Serial Connections: - Router1 S0/0/0 ↔ Router2 S0/0/0 - Router2 S0/0/1 ↔ Router3 S0/0/0

Port Assignments: - Switch ↔ Router ports: trunk - PC & Server ports: access - Serial links: one side DCE (requires clock rate)



IP Addressing Plan

VLAN Networks

Floor	VLAN	Network	Subnet	Gateway
Floor1	10	192.168.10.0	255.255.255.0	192.168.10.1
Floor2	20	192.168.20.0	255.255.255.0	192.168.20.1
Floor3	30	192.168.30.0	255.255.255.0	192.168.30.1

Serial Links

Link	IP / Subnet
Router1 ↔ Router2	10.0.0.1 / 10.0.0.2, 255.255.255.252
Router2 ↔ Router3	10.0.0.5 / 10.0.0.6, 255.255.255.252

Servers

Server	IP	Subnet	Gateway
DNS-SERVER	192.168.20.2	255.255.255.0	192.168.20.1
WEB-SERVER	192.168.20.3	255.255.255.0	192.168.20.1

Switch Configuration

Switch1 – Floor1:

```
enable
configure terminal
vlan 10
name FLOOR1
exit
interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit
interface fa0/24
switchport mode trunk
exit
```

- Fa0/1-3: connected to PCs
- Fa0/24: connected to Router1 (trunk)

Switch2 – Floor2:

```
enable
configure terminal
vlan 20
name FLOOR2
exit
interface range fa0/1-3
switchport mode access
switchport access vlan 20
exit
interface fa0/5
switchport mode access
switchport access vlan 20
exit
interface fa0/6
switchport mode access
switchport access vlan 20
exit
interface fa0/24
```

```
switchport mode trunk
exit
```

- Fa0/1-3: PCs
- Fa0/5: DNS-SERVER
- Fa0/6: WEB-SERVER
- Fa0/24: Router2 trunk

Switch3 – Floor3:

```
enable
configure terminal
vlan 30
name FLOOR3
exit
interface range fa0/1-3
switchport mode access
switchport access vlan 30
exit
interface fa0/24
switchport mode trunk
exit
```

- Fa0/1-3: PCs
- Fa0/24: Router3 trunk

Router Configuration (Router-on-a-Stick)

Router1:

```
enable
configure terminal
interface g0/0
no shutdown
exit
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
interface s0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

Router2:

```
enable
configure terminal
interface g0/0
no shutdown
exit
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
interface s0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
interface s0/0/1
ip address 10.0.0.5 255.255.255.252
clock rate 64000
no shutdown
exit
```

Router3:

```
enable
configure terminal
interface g0/0
no shutdown
exit
interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit
interface s0/0/0
ip address 10.0.0.6 255.255.255.252
no shutdown
exit
```

DHCP Configuration

Router1 – Floor1:

```
ip dhcp pool FLOOR1
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.20.2
```

Router2 – Floor2:

```
ip dhcp pool FLOOR2
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 192.168.20.2
```

Router3 – Floor3:

```
ip dhcp pool FLOOR3
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 192.168.20.2
```

Server Configuration

DNS-SERVER

- IP: 192.168.20.2,
- Subnet: 255.255.255.0,
- Gateway: 192.168.20.1
- DNS service enabled
- Record added: www.mycompany.com → 192.168.20.3

WEB-SERVER

- IP: 192.168.20.3,
- Subnet: 255.255.255.0,
- Gateway: 192.168.20.1
- HTTP service enabled
- Homepage content: “Welcome to Multi-Floor Network”

OSPF Configuration

Router1:

```
router ospf 1
network 192.168.10.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

Router2:

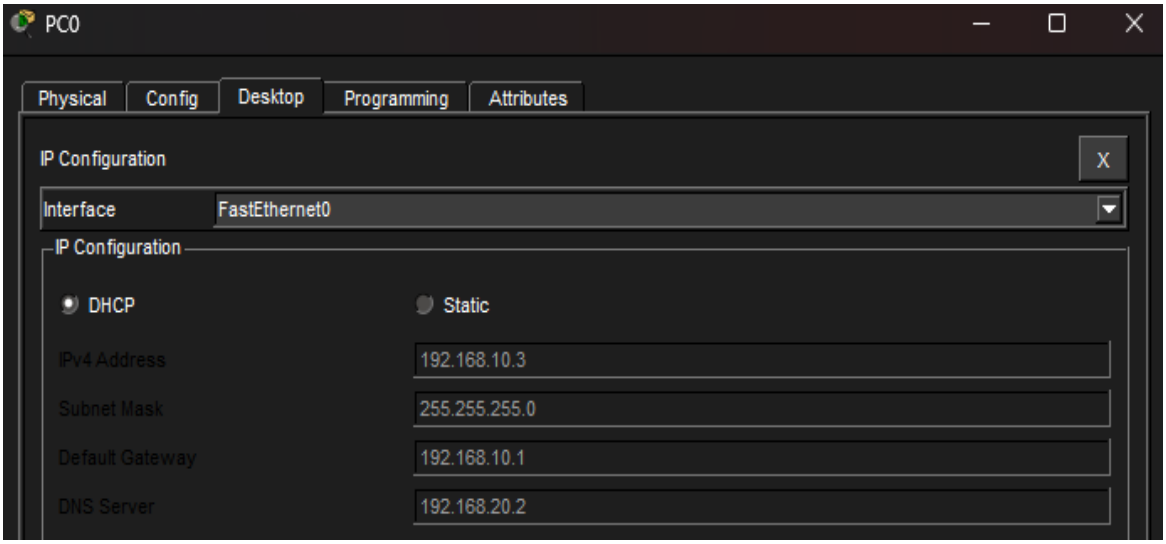
```
router ospf 1
network 192.168.20.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
network 10.0.0.4 0.0.0.3 area 0
```

Router3:

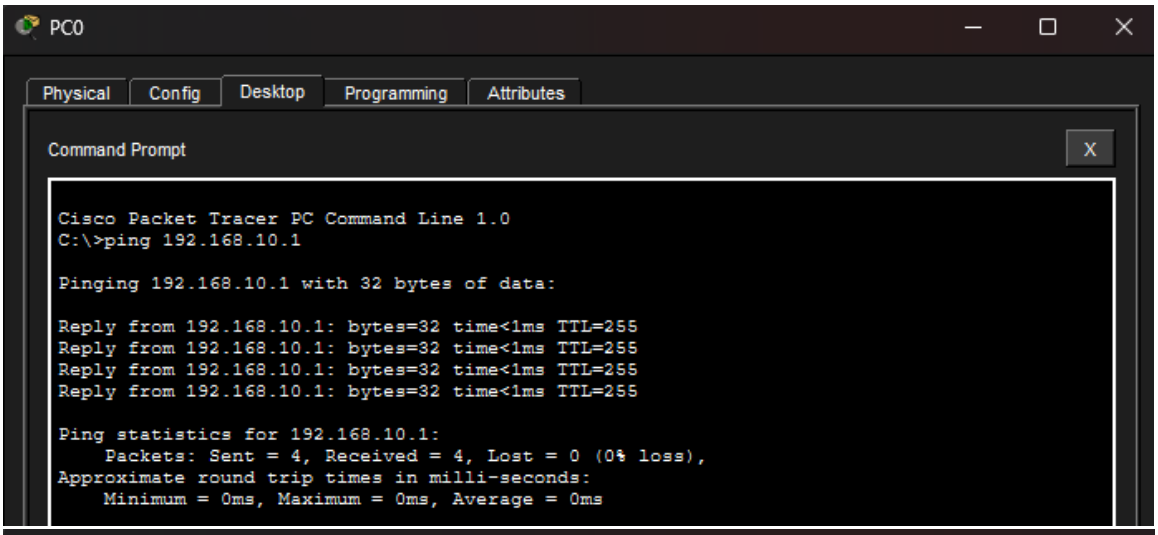
```
router ospf 1
network 192.168.30.0 0.0.0.255 area 0
network 10.0.0.4 0.0.0.3 area 0
```

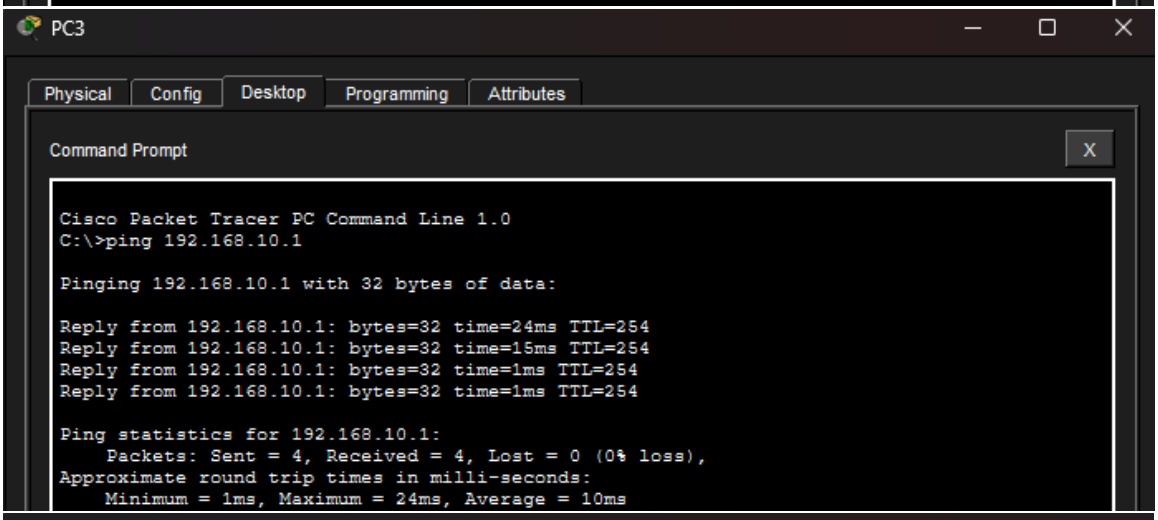
Testing and Verification

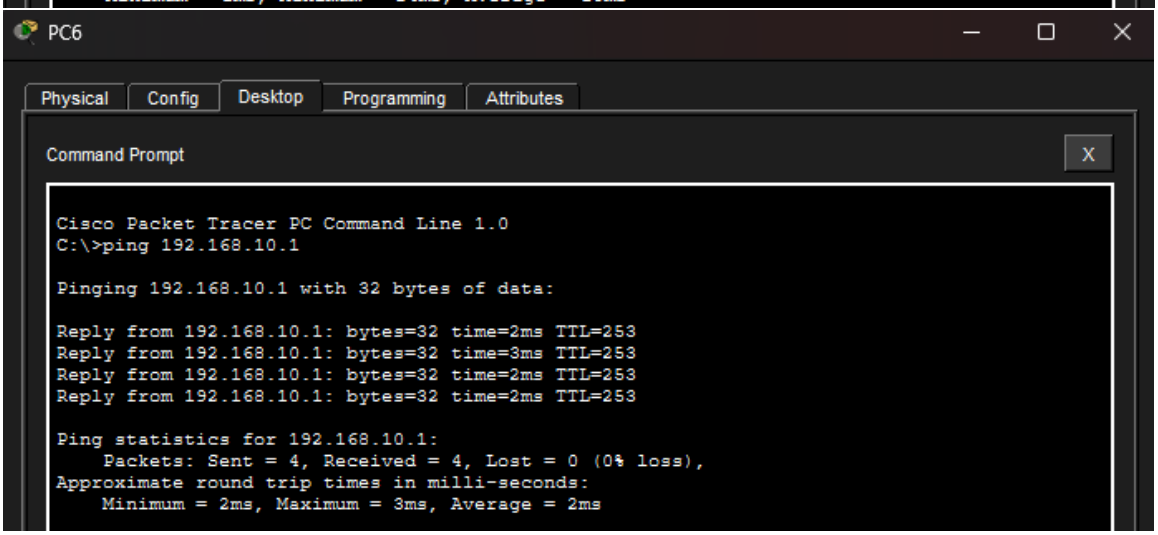
Step 1: Check DHCP on PCs



Step 2: Ping Default Gateway







Step 3: Ping Between Floors

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Step 4: Ping DNS Server

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=31ms TTL=126
Reply from 192.168.20.2: bytes=32 time=2ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 31ms, Average = 9ms
```

Step 5: Ping Web Server

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=1ms TTL=126
Reply from 192.168.20.3: bytes=32 time=11ms TTL=126
Reply from 192.168.20.3: bytes=32 time=1ms TTL=126
Reply from 192.168.20.3: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms
```

Step 6: Test DNS

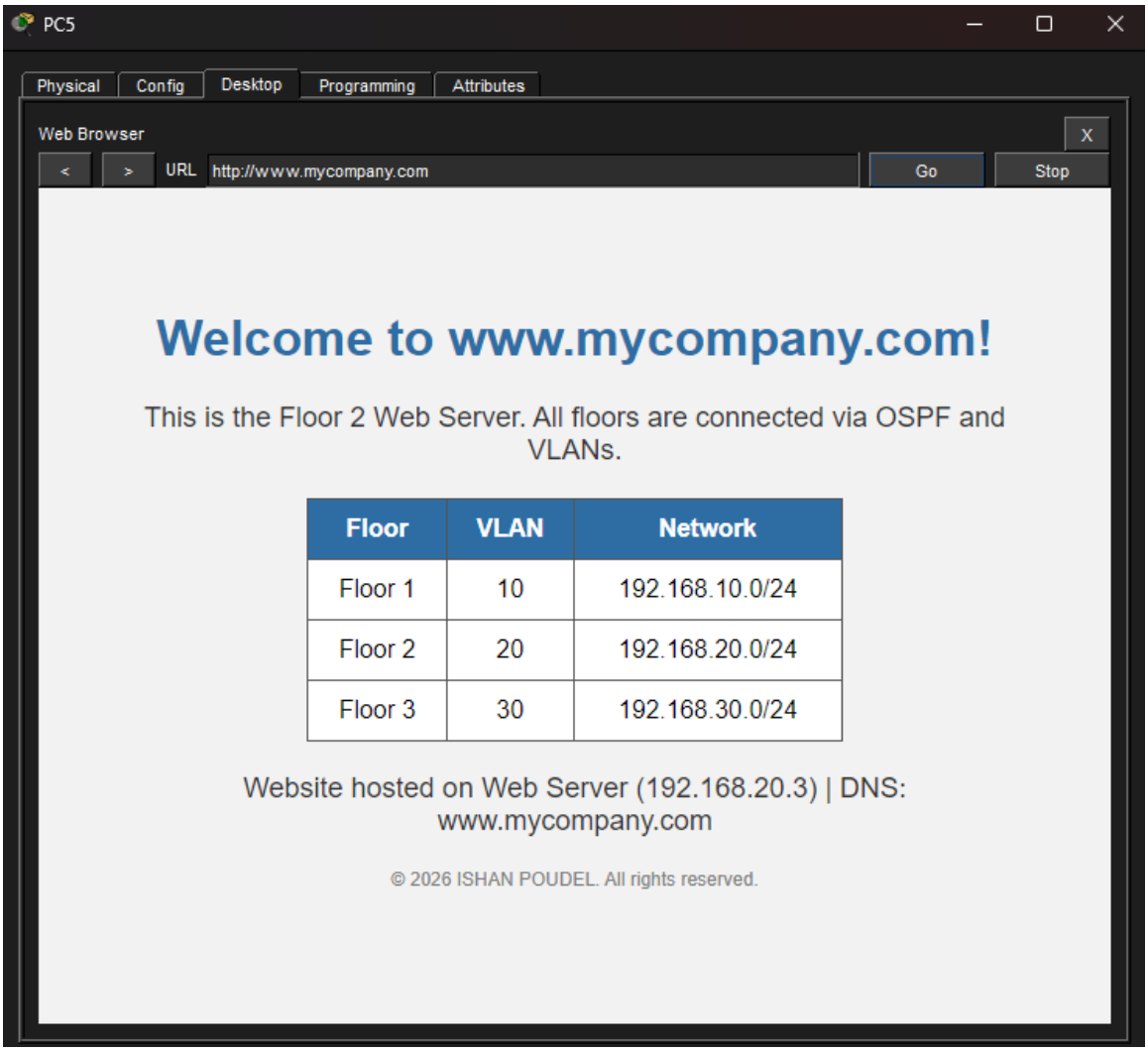
```
C:\>ping www.mycompany.com

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=10ms TTL=126
Reply from 192.168.20.3: bytes=32 time=10ms TTL=126
Reply from 192.168.20.3: bytes=32 time=2ms TTL=126
Reply from 192.168.20.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 5ms
```

Step 7: Open Website



Step 8: Check OSPF Routing

In CLI:

enable
show ip route

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.2, 00:02:54, Serial0/0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
O    192.168.20.0/24 [110/65] via 10.0.0.2, 00:02:54, Serial0/0/0
O    192.168.30.0/24 [110/129] via 10.0.0.2, 00:02:54, Serial0/0/0

Router#
```

Then check neighbors:

show ip ospf neighbor

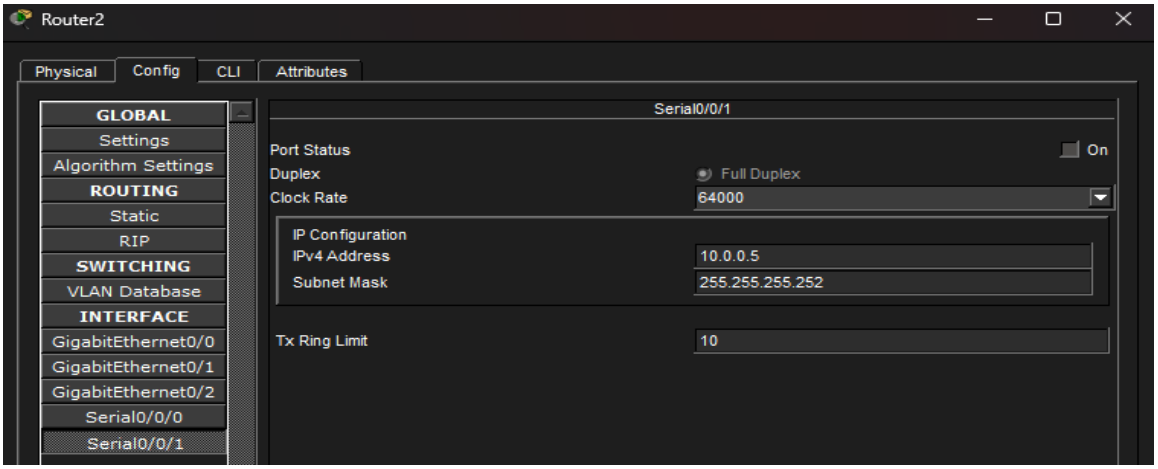
```
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.20.1      0   FULL/  -        00:00:31    10.0.0.2        Serial0/0/0
Router#
```

Step 9: Test Link Failure

1. Disconnect cable between:
   Router2 ↔ Router3

2. Wait few seconds



Now test from PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

After re-connecting cable between R2 and R3:

```
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 2ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253
Reply from 192.168.30.1: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Discussion

This project demonstrates the implementation of a **multi-floor network** using Cisco Packet Tracer. By configuring VLANs and Router-on-a-Stick, we were able to segment network traffic per floor while maintaining interconnectivity. The use of DHCP streamlined IP assignment, reducing manual configuration errors. Implementing a central DNS server simplified name resolution for all devices, allowing users to access the website using `www.mycompany.com` instead of remembering IP addresses. OSPF provided dynamic routing between floors and ensured that changes in network topology, such as link failures, were efficiently handled. Throughout the project, careful attention was given to cabling types (DCE/DTE), trunk/access ports, and IP addressing schemes to ensure reliable communication.

Challenges included ensuring correct sub-interface configurations for VLANs and verifying that all PCs received the correct DHCP information. Troubleshooting these steps provided valuable hands-on experience in network diagnostics and management.

Conclusion

The project successfully achieved the goal of building a fully functional multi-floor network. Key outcomes include:

- Properly segmented VLANs per floor with inter-VLAN routing.
- DHCP servers automatically assigning IP addresses to all PCs.
- Central DNS and Web Server providing network services and web access.
- OSPF ensuring dynamic and reliable routing across all floors.
- Hands-on experience in network troubleshooting and configuration verification.

This project illustrates practical implementation of enterprise-level networking concepts, including VLANs, DHCP, DNS, HTTP, and OSPF, providing a strong foundation for real-world network design and administration.